

ALL-IN-ONE THESIS PROPOSAL & SYSTEM DEFENSE MASTER GUIDE

Notre Dame of Midsayap College • College of Information Technology and Engineering (CITE)

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Approved Title: *Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts* (ForgeGuard System)

SECTION 1: THE PLAIN-ENGLISH TRANSLATOR (SCIENTIST WORDS EXPLAINED)

When speaking to the panel, you will hear and use heavy scientific terminology. Here is what each term actually means in everyday language so you never get confused:

Scientific Term	Plain-English Meaning	Real-World Analogy / Why It Matters
Error Level Analysis (ELA)	A mathematical test that re-saves a picture as a JPEG and looks at what changes.	The Photocopy Analogy: If you photocopy a clean document, the whole page fades equally. But if someone pasted a new number onto it and photocopied it, the pasted number looks blurry or stands out. ELA does that for digital pixels.
Quantization / Compression Error	The microscopic data thrown away when saving an image to make the file size smaller.	GCash compresses a genuine receipt once. A scammer opens it in Photoshop, modifies it, and saves it again. That second save is <i>double compression</i> —leaving a visible scar in the math.
Convolutional Neural Network (CNN)	An AI model designed specifically to scan images using sliding filters.	The Inspector Analogy: A digital detective sliding a magnifying glass across the receipt pixel-by-pixel to detect abnormal tampering edges.
Basic CNN (~2.1M params)	A custom 3-layer sequential neural network.	The Quick Scout: It is simple and shallow, so it looks directly at raw pixel noise without overthinking. It won the benchmark because ELA noise is simple texture.
MobileNetV2 (~3.4M params)	An ultra-efficient neural network designed for smartphones.	Depthwise Separable: It splits heavy math into two quick steps, saving battery and RAM so it can run fast on cheap phones without a GPU.
ResNet50 (~23.5M params)	A massive 50-layer deep network with residual skip connections.	The Overthinker: Built to detect complex objects (dogs, cars). On simple ELA noise, it over-analyzes routine background pixels and mistakes normal noise for tampering.
Sigmoid Activation (≥ 0.50)	The final equation that turns neural math into a percentage between 0% and 100%.	If score $< 0.50 \rightarrow$ Authentic . If score $\geq 0.50 \rightarrow$ Forged . It is an unbiased mathematical coin flip boundary.
True Positive (TP)	A forged receipt correctly flagged as forged.	The scammer's fake receipt was caught and blocked.
True Negative (TN)	An authentic receipt correctly confirmed as authentic.	An honest customer's real receipt was verified.
False Positive (FP)	An authentic receipt wrongly accused of being fake.	A false alarm. An honest customer is embarrassed, but no money is lost.
False Negative (FN)	A forged receipt that slipped through and was marked authentic.	THE WORST ERROR: The merchant releases goods without receiving payment, causing direct financial loss.
Recall (Sensitivity)	The percentage of all fake receipts that the system successfully caught.	Formula: $TP / (TP + FN)$. In cybersecurity, Recall is #1 because letting even one fake slip through means stolen merchandise.
End-to-End Latency	Total time in milliseconds from reading the file to giving the final verdict.	Includes reading file, JPEG decoding, ELA generation, and AI prediction. Basic CNN takes 22.51 ms (less than 1/40th of a second).
Memory Load (Peak RSS)	The maximum amount of RAM the program consumes while running.	Measured in Megabytes. Proves whether the system can run on a cheap merchant laptop without freezing.
Heavily Compressed Condition	Images re-saved with aggressive lossy compression like Facebook Messenger.	Simulates real life: customers send screenshots via Messenger/Viber. We must prove the AI still works when compressed.
Two-Way (3x2) Repeated-Measures ANOVA	A formal statistical test checking if differences between models and conditions are real or just luck.	3 architectures (Basic, MobileNet, ResNet) crossed with 2 conditions (Original vs Compressed). If p-value < 0.05 , the difference is statistically significant.

SECTION 2: PROPOSAL MANUSCRIPT GROUND TRUTH (CHAPTERS 1 & 2)

1. Statement of the Problem (SOP 1 to 5)

Your defense is structured around answering these exact five questions (Section 1.2, pages 7-8):

SOP	Research Question & Purpose	Data Required	Statistical Test
SOP 1	Standard Metrics: Precision, Recall, and F1-score for each model.	Confusion matrix counts (TP, TN, FP, FN) across 5 test seeds.	Descriptive statistics: Mean, SD, Percentages.
SOP 2	Forgery Subtype Accuracy: Accuracy in detecting (2.1) Digitally edited vs (2.2) Programmatically generated fakes.	Separate confusion matrices for digital edits vs template generations.	Subtype accuracy comparisons.
SOP 3	Processing Efficiency & Memory Load: Is there a significant difference in latency and peak RAM among the models? (H ₀ 3)	End-to-end processing time (ms) and peak Resident Set Size (MB) across 5 seeds.	Repeated-Measures ANOVA or non-parametric Friedman test under Bonferroni correction.
SOP 4	Cross-Condition Compression Impact: Is there a significant difference in accuracy between Original High-Res and Heavily Compressed receipts? (H ₀ 1, H ₀ 2)	Accuracy, Precision, Recall, and F1 across both image conditions.	Two-way (3x2) Repeated-Measures ANOVA testing architecture, image condition, and their interaction.
SOP 5	Practical Trade-Off: Which architecture offers the most practical balance of accuracy and efficiency on merchant-grade hardware?	Mean values of accuracy, F1-score, latency, and memory consumption.	Normalized Composite Scoring and Pareto ranking.

2. Scope & Delimitations (Section 1.4) - The Critical Defense Point

Why Downloadable Receipts ONLY (and NOT Mobile Screenshots)?

In GCash, tapping 'Download' generates an official standalone JPEG transaction slip. When users take full phone screenshots instead, the image includes the phone's battery status, carrier logo, and mobile UI chrome. These extraneous elements carry phone-dependent compression artifacts unrelated to GCash. Section 1.4 delimits the study to officially downloaded receipts to ensure controlled, platform-specific forensic rigor.

3. The Expert Validation Controversy: CCJE vs. CITE Faculty

Question from Adviser/Panel Notes: 'Can CCJE students validate? Is that methodologically valid?'

The Dual-Validation Academic Defense:

- Technical Instrument Validation (CITE Faculty):** Section 2.2 explicitly requires three NDMC CITE/Computing faculty members (independent of your defense panel) to validate the computational pipeline, ELA algorithms, CNN tensor dimensions, and ANOVA statistical tools. CCJE students cannot validate Computer Science code.
- Document Realism / Face Validity (CCJE Faculty & Students):** In criminology and forensic science, CCJE researchers specialize in **Questioned Document Examination (QDE)**. Utilizing CCJE students or faculty to rate the visual realism of the forged receipts using your 4-point rubric (layout fidelity, font consistency, absence of crude editing scars) is completely valid. In fact, proving that trained criminology students cannot visually distinguish the fakes reinforces why an automated AI tool is urgently needed!

4. Empty Result Tables for Thesis Chapters 3 & 4

Table for SOP 3: End-to-End Latency and Memory Load (Fixed Merchant PC Benchmark)

Architecture	Condition	End-to-End Latency (ms) Mean ± SD	Peak Memory (MB) Mean ± SD	Params	Statistical Test (F / χ^2)	p-value	Decision (H ₀ 3)
Basic CNN	Original	[]	[]	~2.1M	—	—	—
Basic CNN	Compressed	[]	[]	~2.1M	—	—	—
MobileNetV2	Original	[]	[]	~3.4M	—	—	—
MobileNetV2	Compressed	[]	[]	~3.4M	—	—	—
ResNet50	Original	[]	[]	~23.5M	—	—	—
ResNet50	Compressed	[]	[]	~23.5M	—	—	—

SECTION 3: FORGEGUARD SYSTEM GUIDE & NEURAL MECHANICS

1. How ForgeGuard Operates (Step-by-Step Architecture)

ForgeGuard runs a five-tier forensic pipeline entirely inside Python and Streamlit:

- Step 1: Receipt Ingestion & RGB Normalization:** The user uploads a downloadable GCash receipt (PNG/JPEG) or selects a demo sample. The image is normalized to RGB mode.
- Step 2: Error Level Analysis (ELA) Extraction:** The image is re-encoded in memory as a JPEG at calibrated quality Q=90. The pixel difference between original and resaved is amplified by 15.0x. Spliced regions with double compression exhibit distinct error variance.
- Step 3: Thermal Heatmap Generation:** The pixel variance above the noise floor is mapped into a dynamic thermal gradient. Low-noise background pixels remain clean and untinted, while tampered amounts and text glow vivid yellow-red.
- Step 4: Parallel CNN Inference:** The ELA matrix is resized to 128x128x3 and normalized to [0, 1]. It is fed simultaneously to Basic CNN, MobileNetV2, and ResNet50. Each model computes an independent sigmoid verdict (< 0.50 Authentic, ≥ 0.50 Forged) and millisecond latency.
- Step 5: Visual Presentation:** The UI displays the synchronized Tri-Spectral Evidence Decomposition, multi-model confidence cards, live signal telemetry, and the Benchmark Suite.

2. Detailed Comparison of the Three CNN Models

Model Architecture	Design & Parameters	Inference Latency	Standard Acc	Compressed Acc	Empirical Behavior / Why It Happened
Basic CNN	Custom 3-block sequential CNN. ~2.1M parameters.	22.51 ms (Fastest)	92.92%	99.56%	Highest Accuracy Winner: Because ELA noise is low-level spatial texture, the shallow layers capture the raw edges directly without distorting them.
MobileNetV2	Inverted residual depthwise separable CNN. ~3.4M parameters.	215.75 ms	86.06%	93.42%	Optimal Edge Balance: Highly efficient on mobile hardware. Achieves excellent generalization across both uncompressed and compressed conditions.
ResNet50	50-layer deep residual network with skip shortcuts. ~23.5M parameters.	395.03 ms (Heaviest: 638MB RAM)	53.54%	53.95%	The Over-fitting Trap: ResNet50's 50 layers are designed for semantic objects. On subtle noise maps, it over-analyzes background variance, classifying all images as forged.

SECTION 4: TURNKEY PRESENTATION & LIVE DEMO SCRIPT

Follow this script turn-by-turn during your oral presentation and live system demonstration:

PART 1: FORMAL OPENING (Slide 1)

WHAT TO DO: Stand straight, look at the panel. Daniela and Rogie stand side by side.

WHAT TO SAY: "Good morning, members of the panel, our thesis adviser Ms. Doris Ann Mariano, and guests. We are Daniela Ungab and Rogie Bacanto, fourth-year Computer Science students at Notre Dame of Midsayap College. Today, we present our thesis proposal: *Receipt or Deceit: A Cross-Architecture Analysis of Convolutional Neural Network Models in Detecting Forged Digital Transaction Receipts*, implemented through our forensic system, ForgeGuard."

PART 2: THE PROBLEM & MIDSAYAP CONTEXT (Slides 2-3)

WHAT TO DO: Advance to Slide 2 showing digital payment growth and GCash fraud advisory.

WHAT TO SAY: "In the Philippines, digital payments now account for 52.8% of everyday transactions. In Midsayap, micro-merchants and student entrepreneurs rely on downloadable GCash receipts to verify payments before handing over merchandise. Fraudsters exploit this by using editing software like Photoshop or online receipt generators to change the amount and reference number. Because buyers send these receipts through Messenger, social media compression blurs visual flaws. Our research solves this by evaluating three CNN architectures on Error Level Analysis to build a fast, automated detection tool."

PART 3: LIVE SYSTEM DEMO (Open forgeguard.streamlit.app or localhost)

STEP 1 - LOAD AUTHENTIC SAMPLE: Click the 'Authentic Receipt' button on the Classify screen.

WHAT TO SAY: "We now demonstrate ForgeGuard. When an authentic receipt is evaluated, our ELA engine re-encodes the receipt at Q=90 JPEG quality. Under the ELA tab, notice that the compression noise is completely uniform across the canvas. Both Basic CNN and MobileNetV2 verify the receipt as Authentic with high confidence in under 25 milliseconds."

STEP 2 - LOAD EDITED SAMPLE: Click the 'Edited Tampering' button. Switch to the Tamper Heatmap tab.

WHAT TO SAY: "Next, we load an altered receipt where the transaction amount was tampered with. In the ELA matrix and Tamper Heatmap Overlay, the modified amount glows in warm thermal red because the spliced region underwent double compression. The models detect these residual spikes and classify it as Forged."

STEP 3 - SHOW TRI-SPECTRAL DECOMPOSITION: Scroll down to the Tri-Spectral Gallery.

WHAT TO SAY: "Below the results, our Tri-Spectral Decomposition displays the original document, the raw ELA noise matrix, and the thermal hotspot localization side-by-side. The telemetry bar provides real-time mathematical signals—Mean Noise Energy, Spatial Variance, and Peak Residual—showing exactly how the AI evaluates pixel signals."

STEP 4 - BENCHMARK SUITE: Click 'Model Comparison' in the sidebar.

WHAT TO SAY: "Finally, our Model Benchmark Suite compares the models across both Standard and Compressed conditions. Our empirical results show that the compact Basic CNN achieves 92.92% standard and 99.56% compressed accuracy at just 22.51 ms, proving that simpler, lightweight architectures outperform heavy deep networks for image forensics."

PART 4: TOP 5 TOUGHEST DEFENSE QUESTIONS & WINNING ANSWERS

Q1: Why did you choose ELA instead of training directly on raw receipt images?

Answer: "Raw receipts have varying colors, promotional banners, and fonts that cause neural networks to memorize branding rather than tampering. ELA removes visual cosmetics and exposes the underlying mathematical JPEG compression grid, forcing the AI to detect authentic compression physics."

Q2: Why did ResNet50 perform poorly compared to Basic CNN?

Answer: "ResNet50 has 50 layers designed for semantic object recognition. ELA noise maps do not have semantic objects—only microscopic residual gradients. ResNet50's deep layers over-abstracted this noise and treated normal background compression as tampering, leading to high false alarms."

Q3: What makes this Computer Science instead of an IT Capstone?

Answer: "This study is a quantitative cross-architecture comparative experiment analyzing convolutional feature representations, ELA residual mathematics, and two-way repeated-measures ANOVA statistical modeling under lossy quantization, directly aligned with core CS image forensics."